

Design Verification Plan & Report (virtual test) — t5_r1_flash

Doc No. VBF-T5R1FLASH-DVPR-01 · Rev A (2026-08-25) · Virtual-test version; all results from simulation (run-pyamm / calc-energy), conclusion-grade values DFN.

#	Verification item	Condition	Result	Determination	Source
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1	1C discharge capacity	1C CC discharge 4.7→2.5 V, 25 °C	8.26 Ah	informational (no contract threshold)	cell/r9_lnmo_r6_1c_dfn.json:capacity_ah
2	Energy density	1C discharge energy ÷ contract mass	597.9 Wh/kg	PASS PASS vs ≥500.94	cell/r9_lnmo_r6_energy_dfn.json:energy_density_wh_kg
3	4C fast-charge temperature	4C charge, 45 °C ambient, lumped thermal, DFN	T_max 327.55 K (54.4 °C)	PASS PASS vs ≤333.15 K	cell/r9_lnmo_r6_4c45_dfn.json:T_max_K
4	4C fast-charge plating	same, anode potential series	min +14.7 mV (>0)	PASS PASS (plated=false)	cell/r9_lnmo_r6_4c45_dfn.json:anode_potential_v
5	Voltage window	parameter set limits	2.5–4.7 V	—	LNMO.json
6	4C charge acceptance	protocol cap	2.0 Ah accepted	informational	cell/r9_lnmo_r6_4c45_dfn.json:capacity_ah
7	DC resistance (formula)	calc-energy V ² /4P formula	8.0 mΩ	informational	cell/r9_lnmo_r6_energy_dfn.json:dcr_ohm
8	Nail penetration	—	N/A (beyond pure simulation boundary, requires physical experiment)	—	—
9	Overcharge to thermal runaway	—	N/A (same)	—	—
10	Crush / drop / mechanical	—	N/A (same)	—	—
11	Cycle life	—	N/A (aging model not in scope of this case; no SEI aging protocol run)	—	—
12	Rate-pulse internal resistance	—	N/A (same)	—	—

Conclusion: contract requirements **ACHIEVED** — energy density 597.9 ≥ 500.94 Wh/kg PASS, 4C no-plating PASS (anode min +14.7 mV), T_max 327.6 °C ≤ 60 °C PASS.

Not covered (paper limitations): mechanical abuse, overcharge, cycle life, low temperature, rate pulse — require physical experiments; electrolyte transport values are estimates pending true MD endorsement (real_compute=false).